影像處理實習

Week 3. 幾何變換&強度轉換

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幾何變換

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Part 1.WrapAffine套件介紹

使用公式： (a, b, c, d操作旋轉、縮放、裁切， tx, ty操作平移) 對圖上的每一點做

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

套件用法：

```
cv2.warpAffine(src, M, dsize)
```

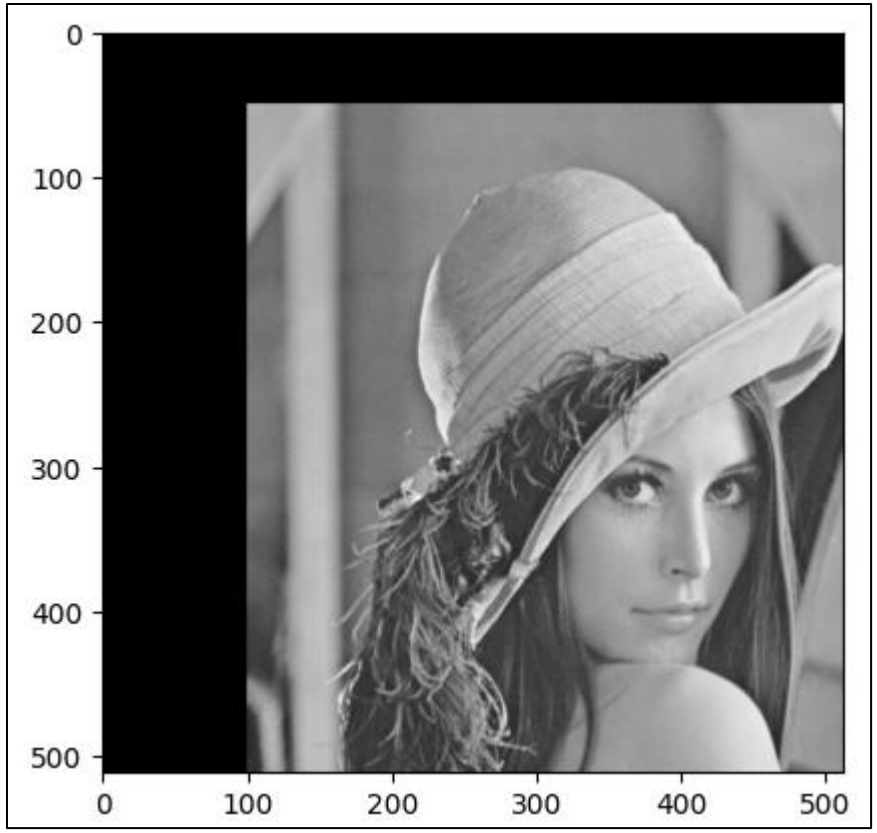
- Scr : 影像路徑
- M：變換矩陣
- Dsize：輸入&輸出影像大小

Part 1.平移(translate)

$$x' = x + t_x$$

$$y' = y + t_y$$

```
h, w = img.shape[:2]
M_translate = np.float32([[1, 0, 100], [0, 1, 50]])  # x 移
動 100, y 移動 50
translated = cv2.warpAffine(img, M_translate, (w, h))
```

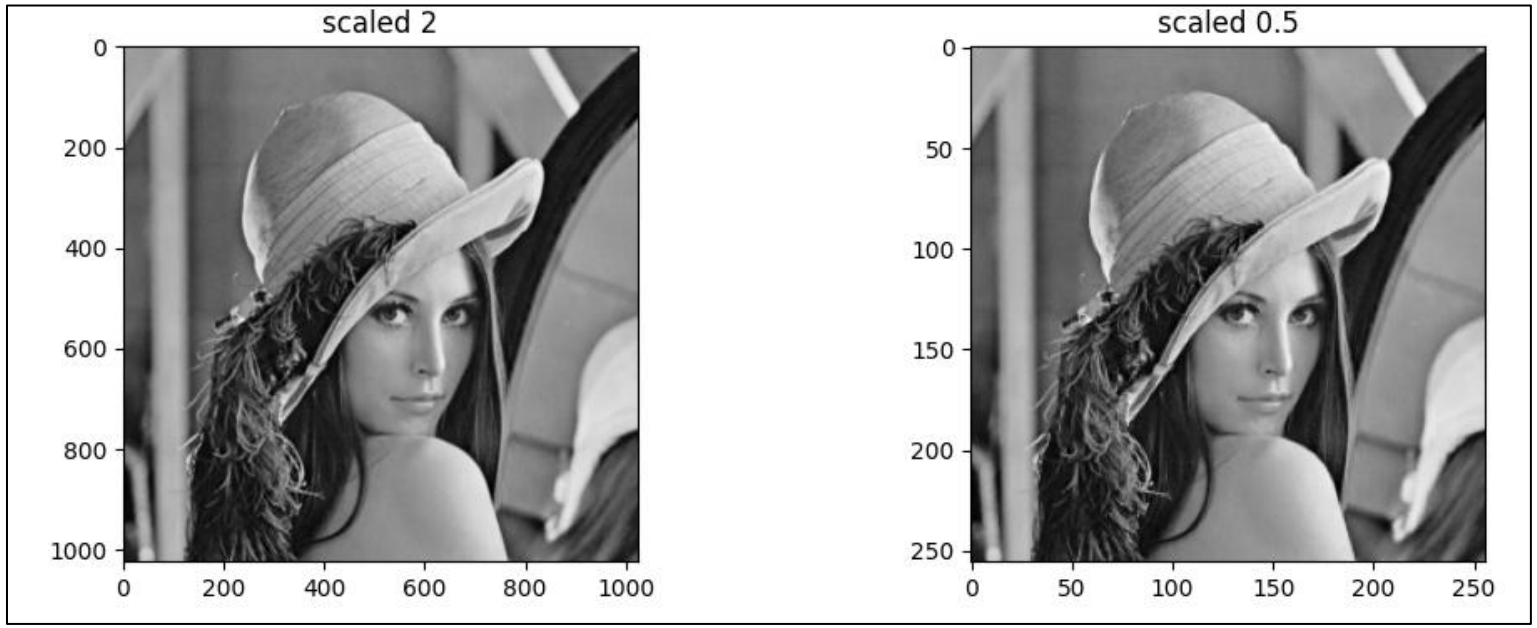


Part 1.縮放(scaled)

$$x' = s_x \cdot x$$

$$y' = s_y \cdot y$$

```
scaled_2 = cv2.resize(img, (w*2, h*2))
scaled_0_5 = cv2.resize(img, (w//2, h//2))
```



Part 1.旋轉(rotated)

原座標旋轉矩陣 (繞原點)：

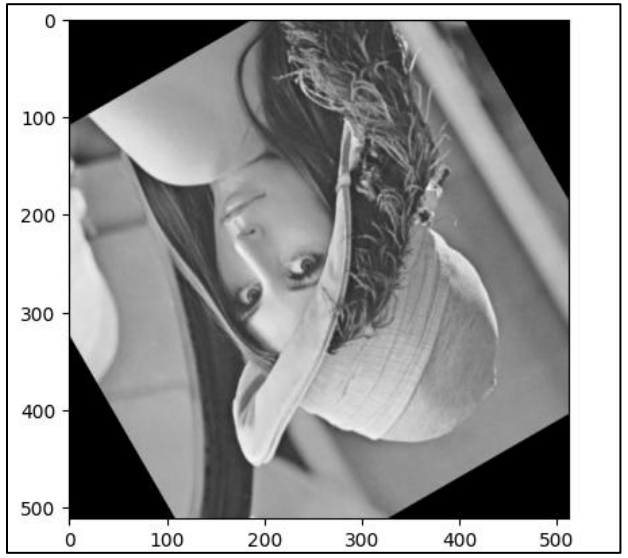
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

```
def rotation_matrix(degree, cx, cy):
  theta = np.radians(degree)  # 角度轉弧度
  cos_t = np.cos(theta)
  sin_t = np.sin(theta)
  R = np.array([[cos_t, -sin_t, (1-cos_t)*cx + sin_t*cy],
                [sin_t, cos_t, (1-cos_t)*cy - sin_t*cx]])
  return R
```

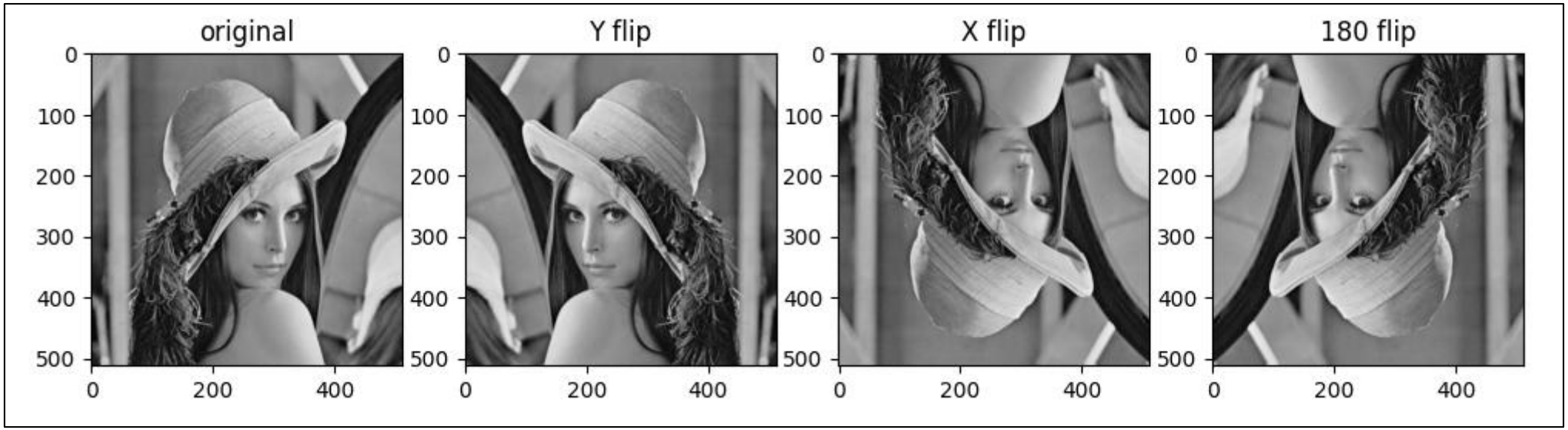
影像版本旋轉矩陣 (繞圖中心)：

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x - cx \\ y - cy \end{bmatrix} + \begin{bmatrix} cx \\ cy \end{bmatrix}$$

1.改變原點座標 -> 2.執行旋轉 -> 3.回歸原本座標



Part 1.鏡射



Part 1.仿射變換

A點 -> B點變換矩陣為?

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

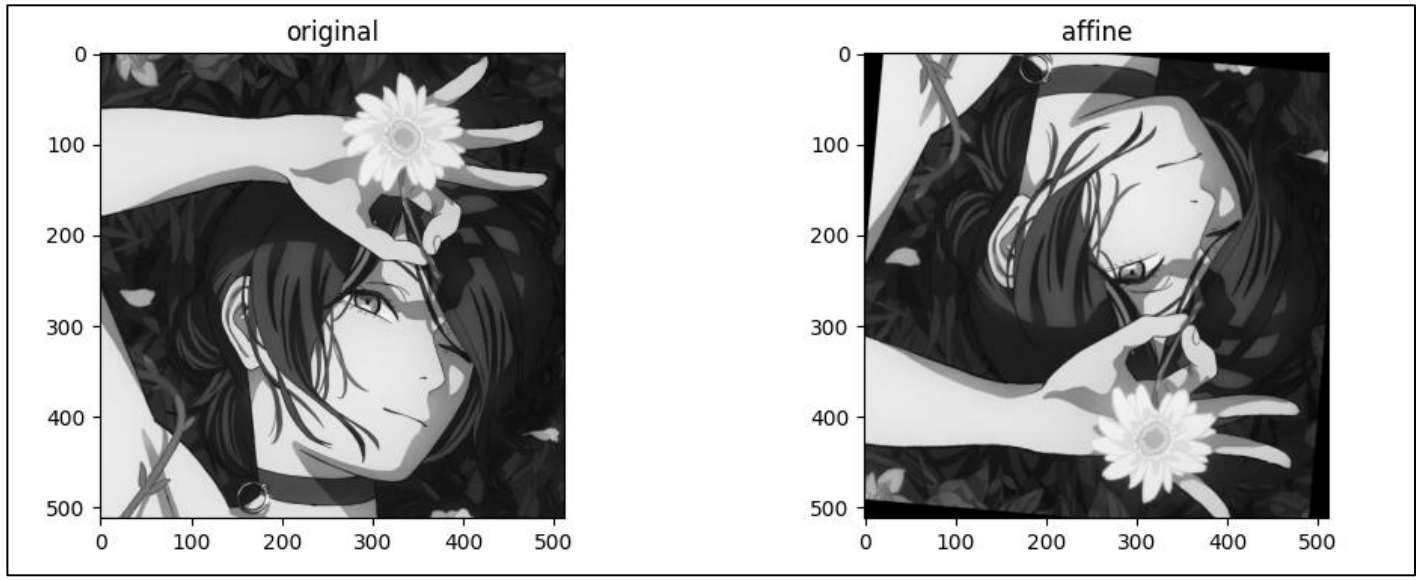
- $x' = a \cdot x + b \cdot y + \mathrm{t_x}$
- $y' = c \cdot x + d \cdot y + \mathrm{t_y}$

$$\begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix} \ ?$$

$$\begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 8 & 2 \\ 6 & 1 \end{bmatrix} \ ?$$

本日作業

1. 找一張圖片
2. 縮放size到512*512
3. 旋轉175度
4. 以y軸鏡射



請繳交1.原圖、2.處理後的圖 (要有圖座標)，檔案限制.jpg或.png